

HK IMO Training 2020-2021
Problem Set 12
For 23 January 2020



Problem 45. Let a, b, c be positive real numbers such that $abc = 1$. Prove that

$$\frac{1}{a^4 + b^3 + c^2} + \frac{1}{b^4 + c^3 + a^2} + \frac{1}{c^4 + a^3 + b^2} \leq 1.$$

Problem 46. Find all real constants c for which there exists a strictly increasing sequence $\{a_n\}_{n \geq 1}$ of positive integers such that

$$\frac{a_{2n-1} + a_{2n}}{a_n} = c$$

for any positive integer n .

Problem 47. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$yf(x) + f(y) \geq f(xy)$$

for any real numbers x and y .

Problem 48. Let $P(x)$ be a polynomial with real coefficients whose (complex) roots are distinct positive real numbers. Is it possible that the polynomial $Q(x) = x(x+1)(x+2)(x+4)P(x) + c$ is the k th power of a polynomial with real coefficients for some positive integer $k \geq 2$ and real number c ?